The University of Lahore

Department of Computer Science & IT

Midterm Examination

Fall-2024

Course Title:	Software Engineering	Course Code:	CS4346	Credit Hours:	3
Course Instructors:	Ms. Saman Asad, Mr. Ali Ilyas	Program Name:	BSCS		
Time Allowed:	**60 Minutes**	**Maximum Marks:**	**30**		
Date:	**25-11-2024**	Course Mentor:	Dr. Atif Ikram		
Student's Name:		Signature:			
Reg. No:		Section:			

Question #1: MCQ's (10 marks)

1.Software is computer programs and associated documentation that includes.

A. Requirements
B. Design Models
C. User Manuals
D. All of these.

2. Generic softwares are developed to be sold to a range of different customers.

A. True
B. False

3. Tools to support the early process activities of requirements and design are.

A. Lower CASE.
B. Upper CASE.
C. CASE Tools.
D. None of these.

4. Which of the following is attribute of good software?

A. Maintainable
B. Efficient
C. Acceptable
D. Dependable
E. All given options

5. Costs vary based on the system's type that being developed and the system requirement attributes such as performance and system reliability?

A. True
B. False

6. The main goal of agile methods is to reduce overheads in the software process by limiting ... ?

A. Design
B. Code
C. Documentation
D. All given options

7. The process of programming team look for possible software improvements is called ...?

A. Pair programming
B. Refactoring
C. Code testing
D. All given options

8. Usability is not a Testable non-functional requirement?

A. True
B. False

9. __________ is intended to show that a system conforms to its specification and meets the requirements of the system customer.

A. Requirements Specification
B. Validation
C. Verification
D. Both (b) and (c)

10. The functionality of services the system should provide and how the system should react to particular inputs is?

A. Functional Requirements
B. Non-Functional Requirements
C. Functional and Non-Functional Requirements
D. None of the given options

The University of Lahore
Department of Computer Science & IT
Midterm Examination

Fall-2024

Course Title:	Software Engineering	Course Code:	CS2309	Credit Hours:	3
Course Instructors:	Ms. Saman Asad, Mr. Ali Ilyas	Program Name:	BSCS		
Time Allowed:	**60 Minutes**	**Maximum Marks:**	**20**		
Date:	**25-11-2024**	Course Mentor:	Dr. Atif Ikram		
Student's Name:		Signature:			
Reg. No:		Section:			

Subjective Part

Question #1: **5 Marks**

Why is it essential to conduct formal verification and validation processes in critical systems (e.g., medical software, aviation systems)? Discuss the implications of skipping these steps

Solution:

Verification and validation (V&V) ensure that a system meets its specified requirements and operates correctly. In critical systems, such as medical software or aviation systems, these processes are vital because:

1. **Error Elimination:** Early detection and correction of errors reduce the risk of catastrophic failures (e.g., incorrect drug dosage in medical software).
2. **Safety and Reliability:** These systems often operate in life-critical environments where failures can lead to loss of life or severe damage.
3. **Compliance and Standards:** Many industries are regulated by laws and standards (e.g., FDA for medical devices), which mandate rigorous V&V.

Implications of Skipping V&V:

1. **Safety Risks:** Undetected bugs can lead to accidents or fatalities.
2. **Legal Consequences:** Organizations may face lawsuits or fines for non-compliance with regulations.
3. **Reputational Damage:** A system failure in critical systems harms the organization's credibility and trustworthiness.
4. **Higher Costs:** Fixing errors post-deployment is more expensive than addressing them during development.

Question #2: **5 Marks**

For the following systems, recommend the most appropriate software development life cycle model, and justify your choices:

1. A weather forecasting system requiring real-time data processing and reliability.
2. A mobile payment application targeting seamless and secure transactions.
3. A student management portal for a small educational institution.

Solution:

1. **Weather Forecasting System:**
   Model: Spiral Model
   Justification:
   - The system requires constant updates and testing due to real-time data processing and evolving algorithms.
   - The iterative nature of the Spiral Model allows risk assessment at each cycle, ensuring reliability and accuracy.
2. **Mobile Payment Application:**
   Model: Incremental Model
   Justification:
   - Core functionalities like secure payments can be developed and tested early, while additional features like loyalty rewards can be added incrementally.
   - Frequent releases ensure user feedback for better usability and security updates.
3. **Student Management Portal:**
   Model: Waterfall Model
   Justification:
   - This system has well-defined requirements and a predictable workflow, making the sequential approach of the Waterfall Model suitable.
   - It simplifies management for a small institution with limited budget and technical expertise.

Question #3: 10 Marks

Scenario:

GreenTech Solutions is an organization focused on developing an eco-friendly logistics management system. This system aims to optimize delivery routes to reduce fuel consumption and carbon emissions while ensuring timely deliveries. GreenTech Solutions caters to:

1. **Large Logistics Providers:** These companies require advanced analytics to track multiple deliveries across regions and minimize operational costs.
2. **Small Businesses:** Local stores seeking simple yet effective solutions to manage their deliveries and monitor environmental impact.
3. **Environmental Agencies:** Organizations interested in accessing detailed reports on carbon footprint reduction for regulatory and promotional purposes.

Task:

Using the template provided, define two user requirements for the logistics management system.

Requirement Template:

<u>Solution:</u>

Requirement 1

Function	Route Optimization
Description	Calculate and suggest the most fuel-efficient delivery routes.
Inputs	Delivery locations, vehicle type, current traffic data.
Source	Logistics manager, traffic API.
Outputs	Optimized route plan.
Destination	Delivery driver's navigation system.
Action	Minimize fuel consumption and delivery time.
Requirements	1. The system must interface with live traffic APIs for real-time data. 2. The system must consider vehicle-specific parameters (e.g., fuel efficiency) when generating routes. 3. The interface must allow manual override for routes in exceptional circumstances.
Pre-Conditions	Delivery locations must be specified, and traffic API must be accessible.
Post-Conditions	Route plan is saved and communicated to the driver.
Side-Effects	Potential reliance on inaccurate traffic data.

Requirement 2

Function	Carbon Emission Reporting
Description	Generate detailed reports on emissions for each delivery route.
Inputs	Distance traveled, fuel type, and quantity consumed.
Source	Vehicle tracking system and driver logs.
Outputs	Carbon emission report (in kilograms of CO2).
Destination	Environmental agency portal and company database.
Action	Provide actionable insights for reducing emissions.
Requirements	1. The system must adhere to industry-standard calculations for carbon emissions. 2. The reports must be exportable in multiple formats (e.g., PDF, Excel). 3. The system must provide cumulative monthly and annual emission summaries.
Pre-Conditions	Vehicles must report accurate fuel consumption data.
Post-Conditions	Report is saved and available for review.
Side-Effects	Additional computational resources may be required for generating reports.

Good Luck